

SOFTWARE REQUIREMENTS SPECIFICATION

AI RESUME ASSISTANT

"Scan. Match. Improve."

Date: September 8, 2026

Team Size: 4 Members

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1. INTRODUCTION

1.1 Purpose

This Software Requirements Specification (SRS) document defines the requirements for the AI Resume Assistant application. The purpose of this document is to provide a comprehensive description of the system's functionality, constraints, and interfaces to guide the development team through implementation.

1.2 Project Overview

The AI Resume Assistant is a web-based application designed to help college students and early-career job seekers optimize their resumes for specific job opportunities. The application uses intelligent keyword matching and AI-powered text enhancement to provide users with actionable recommendations for improving their resumes.

1.3 Scope

The AI Resume Assistant will include:

- User authentication and account management
- Resume file upload and storage (PDF and DOCX)
- Automated resume content parsing and extraction
- Job posting input via URL or copy-paste
- Resume-to-job matching with ATS-style scoring
- AI-powered resume bullet point improvement with validation
- Updated resume export as PDF

1.4 Intended Users

The primary users of this application are:

- College students preparing for internships
- Recent graduates seeking entry-level positions
- Early-career professionals (0–3 years experience)

2. SYSTEM OVERVIEW

2.1 High-Level Features

Resume Upload & Parsing: Users upload resumes in PDF or DOCX format. The system automatically extracts text and parses it into structured sections (contact info, experience, education, skills).

Job Posting Acquisition: Users provide job postings either by pasting a URL (automatic extraction attempted) or copying the job description text. The system parses the job posting into structured data.

ATS-Style Matching: The system compares the resume to the job posting, calculates a match score (0–100%), and identifies matched, missing, and weak keywords.

AI Bullet Enhancement: Users select resume bullets to improve. AI generates 3 alternative versions tailored to the job requirements. Users choose their preferred option.

Resume Export: Users download their updated resume with selected improvements as a PDF file.

2.2 System Architecture

Three-Tier Architecture:

- **Presentation Layer (Frontend):** React web application handling user interface, form inputs, and results display
- **Business Logic Layer (Backend):** Node.js/Express API handling parsing, matching, scoring, and AI orchestration
- **Data Layer:** PostgreSQL database for user data, resumes, jobs, and analyses; file storage for resume documents

2.3 Key Components

Component	Responsibility
Frontend	User interface, form inputs, results display, resume editor
Backend API	Business logic, document parsing, analysis engine, AI orchestration
Database	Persistent storage for users, resumes, jobs, analyses
File Storage	Secure storage of uploaded resume documents
AI Service	API for generating improved resume bullets

3. USER REQUIREMENTS

3.1 User Types

Student/Job Seeker

Primary user of the application. Uploads resume, provides job postings, views analysis results, and improves resume bullets.

Responsibilities: Upload resumes, input job postings, review recommendations, approve improved bullets

3.2 User Needs & Goals

User Need	How It Is Met
Tailor resume quickly	Upload once, compare to multiple jobs; reuse parsed resume data
Understand ATS compatibility	Quantified match score (0–100%) with keyword breakdown
Improve resume without guessing	AI-powered suggestions based on specific job requirements
Control changes for accuracy	User reviews and approves all AI suggestions; no auto-modifications
Export polished resume	Download as PDF with selected improvements

4. FUNCTIONAL REQUIREMENTS

4.1 Authentication & Account Management

FR-1.1: Users must be able to register with email and password

Description: New users can create an account by providing an email address and password. The system validates the email format and enforces password strength requirements (minimum 8 characters, mixed case, numbers).

FR-1.2: Users must be able to log in securely

Description: Users provide email and password. The system authenticates using bcrypt or argon2 and issues a JWT token for subsequent requests. Session timeout occurs after 15 minutes of inactivity.

4.2 Resume Management

FR-2.1: Users must be able to upload resume files

Description: Users upload PDF or DOCX files. The system validates file type and size (max 10 MB). Files are stored securely in local filesystem or cloud storage.

FR-2.2: System must automatically parse uploaded resumes

Description: Upon upload, the system extracts text from PDF/DOCX and parses into sections: contact info, summary, experience, education, skills. Extracted data is normalized and stored in database.

4.3 Job Posting Input & Parsing

FR-3.1: Users must be able to input job postings via URL or text

Description: System provides two input methods: (1) URL input with automatic extraction from LinkedIn, Indeed, Glassdoor, etc.; (2) Copy-paste fallback for job description text. If URL extraction fails, system prompts user to use copy-paste.

FR-3.2: System must parse job postings into structured data

Description: System extracts: job title, company, job description, required skills, responsibilities. Data is normalized and skills are identified.

4.4 Resume-Job Matching & ATS Analysis

FR-4.1: System must calculate ATS-style match score

Description: System compares resume keywords to job keywords. Calculates match percentage (0–100%) based on exact matches, partial matches, and missing skills. Score is displayed prominently on results dashboard.

FR-4.2: System must identify matched, missing, and weak keywords

Description: System categorizes job keywords as: (1) Matched – in resume; (2) Missing – not in resume (critical and secondary); (3) Weak – in resume but low frequency. Results displayed in separate lists on dashboard.

4.5 AI-Powered Bullet Enhancement

FR-5.1: Users must be able to select resume bullets for improvement

Description: Dashboard displays all resume bullets. Users click to select a bullet they wish to improve. System highlights the selected bullet and retrieves relevant job requirements.

FR-5.2: System must generate AI-improved bullet options

Description: System sends to API: original bullet, relevant job requirements, supporting evidence from resume. AI generates 3 improved versions. System validates outputs to prevent hallucination (claiming skills not in original resume).

FR-5.3: Users must be able to select and save improved bullets

Description: Users review 3 options, select preferred version, and save. Selected bullet replaces original in updated resume.

4.6 Resume Export

FR-6.1: Users must be able to export updated resume as PDF

Description: System generates PDF containing original resume content with selected improved bullets inserted. File is downloadable with original formatting preserved where possible.

5. NON-FUNCTIONAL REQUIREMENTS

5.1 Performance

- NFR-1.1: Resume upload processing must complete within 5 seconds
- NFR-1.2: Job URL extraction must complete within 10 seconds (timeout if longer)
- NFR-1.3: ATS score calculation must complete within 3 seconds
- NFR-1.4: AI bullet generation may take up to 15 seconds (expected AI latency)
- NFR-1.5: Dashboard page load must complete within 2 seconds

5.2 Security

- NFR-2.1: All passwords must be hashed using bcrypt or argon2 (never stored plaintext)
- NFR-2.2: All network communication must use HTTPS/TLS
- NFR-2.3: System must prevent SQL injection using parameterized queries
- NFR-2.4: System must prevent XSS attacks through input sanitization
- NFR-2.5: Users can only access their own resumes and analyses

5.3 Usability

- NFR-3.1: User interface must be intuitive for non-technical users
- NFR-3.2: Error messages must be in plain English with actionable guidance
- NFR-3.3: Application must be responsive on mobile and desktop browsers
- NFR-3.4: Help tooltips and in-app guidance must be provided for each feature

5.4 Reliability

- NFR-4.1: System must gracefully handle resume parsing failures and notify users
- NFR-4.2: If job URL extraction fails, system must automatically prompt copy-paste fallback
- NFR-4.3: If AI service is unavailable, system must inform user and allow manual editing

6. USER INTERFACE DESIGN

6.1 Key UI Screens

Screen 1: Login/Registration Page

Purpose: User authentication and account creation

Elements:

- Email input field
- Password input field (with strength indicator)
- "Sign Up" and "Log In" buttons
- "Forgot Password" link
- Form validation with real-time feedback

Screen 2: Resume Upload Dashboard

Purpose: Upload resumes and view existing resumes

Elements:

- Drag-and-drop file upload area
- Upload progress indicator
- List of existing resumes with upload dates
- Delete button for each resume
- Supported format warning (PDF, DOCX only)

Screen 3: Job Posting Input

Purpose: Input job posting via URL or copy-paste

Elements:

- Tab 1: "Paste URL" with input field and "Extract" button
- Tab 2: "Paste Description" with textarea and "Submit" button
- Loading state during extraction
- Error message if extraction fails; prompts to use fallback

Screen 4: Analysis Results Dashboard

Purpose: Display ATS match score and keyword analysis

Elements:

- Large match score display (0–100%) with color indicator (red/yellow/green)
- Matched keywords section (list with count)
- Missing keywords section (critical and secondary lists)
- Weak keywords section (keywords in resume but low emphasis)
- Recommendations section (bulleted list of suggestions)
- Resume bullets listed for selection

Screen 5: Bullet Improvement Interface

Purpose: Review and select improved resume bullets

Elements:

- Original bullet displayed (highlighted)
- 3 AI-generated options (numbered 1, 2, 3)
- Edit buttons for each option (users can refine text)
- Select buttons to choose preferred option
- "Save" button to confirm selection
- Loading state while AI generates options

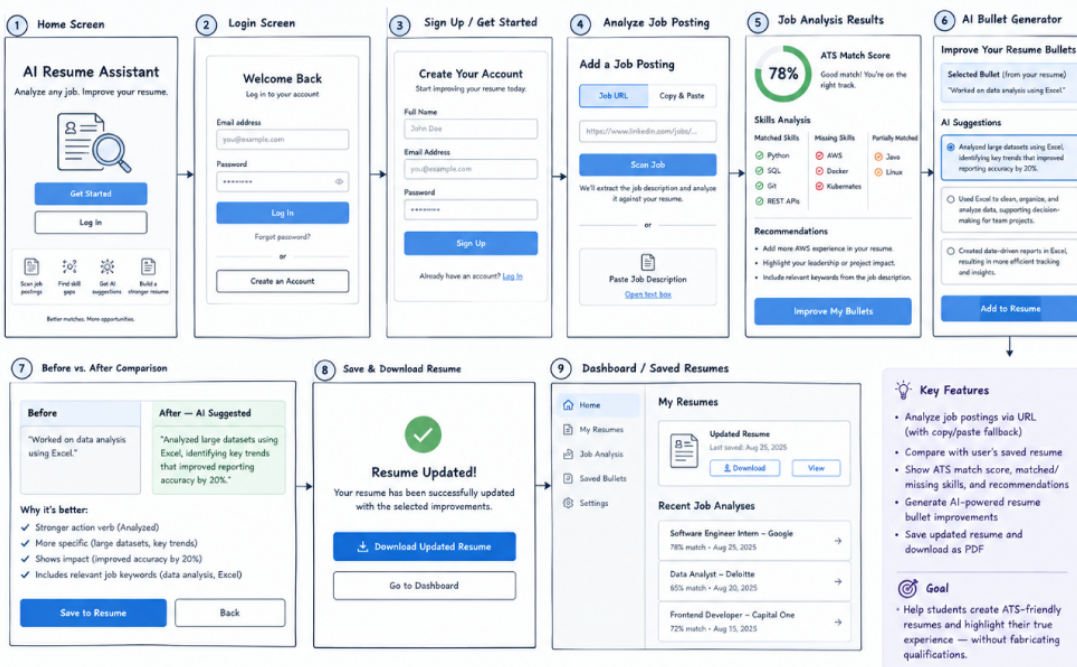
Screen 6: Export/Download Page

Purpose: Export updated resume

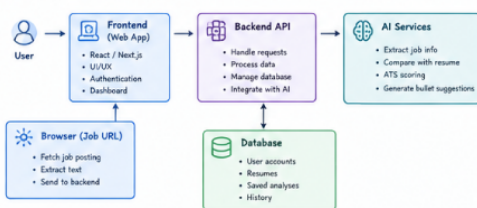
Elements:

- Format selection: PDF (required), DOCX (optional)
- "Download Updated Resume" button
- Preview of resume before download (optional)

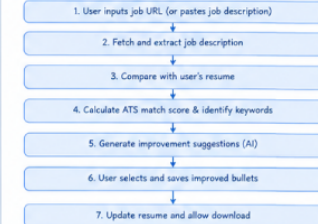
User Flow & Screens (Concept Sketches)



System Architecture



Algorithm / Process Flow



Notes

- If the job URL can't be accessed (e.g., login required or different site structure), show a fallback option to copy and paste the job description.
- Use OpenAI API (or similar) for AI analysis and bullet generation.
- Store user data securely (authentication, encrypted storage).
- Keep the MVP focused on core features: job analysis, ATS matching, and AI bullet generation.

7. SYSTEM CONSTRAINTS

7.1 Technical Constraints

Framework: Application built with Node.js/Express backend and React frontend

Database: PostgreSQL; must support JSON fields for storing parsed data

File Storage: Local filesystem or AWS S3 with maximum 10 MB per file

AI Service: API (requires valid API key and internet connection)

Browser Support: Chrome, Firefox, Safari, Edge (latest versions)

7.2 Environmental Constraints

Development Environment: Windows, macOS, or Linux

Node.js version: 16.x or higher

Hosting: Cloud-based (Render, Heroku, DigitalOcean, or AWS)

7.3 Project Constraints

Timeline: 12-week development cycle

Team Size: 4 members

MVP Target: Complete and deployable by Week 11

No budget for paid services (except AI API costs)

8. GLOSSARY

Term	Definition
ATS	Applicant Tracking System; automated system used by employers to screen resumes based on keywords
API	Application Programming Interface; defines how backend and frontend communicate
Bullet Point	Individual accomplishment or responsibility listed under a job in resume experience section
Keyword	Specific skill, tool, or qualification identified in resume or job posting
Parsing	Automated extraction and organization of structured data from unstructured documents
JWT	JSON Web Token; secure mechanism for transmitting user authentication credentials
Hallucination	When AI generates false or invented information not present in source material
MVP	Minimum Viable Product; smallest set of features providing core value to users

9. REFERENCES

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- [3] React Documentation: <https://react.dev>
- [4] Express.js Documentation: <https://expressjs.com>
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- [6] OWASP Top 10 Web Application Security Risks: <https://owasp.org/www-project-top-ten>

APPROVAL & SIGN-OFF

Role	Name	Signature/Date
Project Lead		
Backend Lead		
Frontend Lead		
QA Lead		